

Deep Learning

Assignment 1

Name: Chandana S | USN: 4GW23CI010

1. What is Deep Learning?

Deep Learning is a subfield of Machine Learning that uses artificial neural networks with multiple layers to automatically learn hierarchical representations of data. Instead of relying on hand-crafted features designed by domain experts, a deep learning model learns the relevant features directly from raw data such as pixels in an image, waveforms in audio, or tokens in text by passing them through successive layers of interconnected neurons.

Each layer in the network transforms its input into a slightly more abstract representation

Deep learning models are typically trained using large labelled or unlabelled datasets, an optimisation algorithm called gradient descent, and a technique called backpropagation that calculates how much each connection in the network contributed to the final error, so that the network's weights can be adjusted to reduce that error over many training iterations. Common architectures include Convolutional Neural Networks (CNNs) for visual data, Recurrent Neural Networks (RNNs) and Transformers for sequential data such as text and speech, and Generative Adversarial Networks (GANs) or Diffusion Models for generating new content.

Deep learning has become dominant in recent years mainly due to three factors: the availability of massive datasets, the growth of computational power (especially GPUs and TPUs), and improved algorithms and architectures.

2. Ten Applications of Deep Learning

- 1 **Computer Vision:** Image classification, object detection, and facial recognition used in security systems, photo tagging, and quality inspection on manufacturing lines.
- 2 **Natural Language Processing (NLP):** Machine translation, chatbots, sentiment analysis, and large language models like ChatGPT that understand and generate human language.
- 3 **Speech Recognition:** Converting spoken language into text for virtual assistants such as Siri, Alexa, and Google Assistant, and for automatic captioning of videos.
- 4 **Autonomous Vehicles:** Self-driving cars use deep learning to perceive their surroundings, detect pedestrians and obstacles, and make real-time driving decisions.
- 5 **Healthcare and Medical Imaging:** Detecting tumours in X-rays, MRIs, and CT scans; predicting disease risk; and assisting in drug discovery and genomics research.
- 6 **Recommendation Systems:** Platforms like Netflix, YouTube, and Amazon use deep learning to predict what content or products a user is likely to enjoy or buy.
- 7 **Fraud Detection:** Banks and payment companies use deep learning to identify unusual transaction patterns that may indicate credit card fraud or money laundering.

- 8 **Generative AI:** Creating realistic images, art, music, and video using GANs and diffusion models
- 9 **Robotics:** Enabling robots to perceive their environment, grasp objects, and learn complex manipulation tasks through reinforcement and deep learning.
- 10 **AI-Designed Computer Chips:** Using deep reinforcement learning to automatically design the physical layout of computer chips, a task traditionally done by human engineers over weeks. This application is explored in detail below.

3. Detailed Application: AI-Designed Computer Chips

3.1 Background

One of the most advanced and less commonly discussed uses of deep learning is chip floorplanning — the process of physically arranging the components of a computer chip (such as memory blocks and logic units) on a piece of silicon. Traditionally, this is one of the most difficult and time-consuming stages of chip design. Expert human engineers can take several weeks to months to produce a good floorplan for a single chip, because the placement has a huge effect on the chip's speed, power consumption, and manufacturing cost, and the number of possible arrangements is astronomically large.

3.2 How Deep Learning Is Applied

Google's AI research team framed chip floorplanning as a reinforcement learning problem, similar in spirit to how AlphaGo learned to play the board game Go. In this framing, the deep learning "agent" views the chip layout as a grid and treats every decision about where to place the next component as a "move." A graph neural network is first used to learn a representation of the chip's netlist — the diagram describing how millions of components need to be electrically connected. The reinforcement learning agent is then trained through repeated trial and error: it proposes a placement, a reward signal is calculated based on estimated wirelength, congestion, timing, and density, and the network's internal weights are updated to favour placements that produce a higher reward. Over many iterations and many training chips, the agent learns a general strategy for producing good layouts, rather than solving each chip from scratch using traditional heuristic algorithms.

3.3 Results and Impact

According to Google's published research, this approach was able to generate chip floorplans in a matter of hours that matched or outperformed layouts produced by human experts over several weeks. The technique, later named AlphaChip, was reportedly used in the design of Google's own Tensor Processing Units (TPUs) — the very type of chips used to train large deep learning models — creating an interesting feedback loop where deep learning helps design the hardware that trains deep learning models. Google has also released trained checkpoints of this system to help other chip designers and researchers build on the approach.

3.4 Why This Is a Significant Application

- It automates a task that requires years of specialised engineering expertise, rather than a task with abundant labelled data such as image tagging.
- It compresses a multi-week manual process into a few hours, directly reducing the time and cost of designing new chips.
- It demonstrates deep learning being applied to its own hardware supply chain, potentially accelerating future improvements in AI computing power.
- It generalises across different chip designs after training, rather than needing to be re-solved from first principles every time, showing the power of learned heuristics over traditional algorithmic search.

3.5 Conclusion

AI-designed computer chips illustrate how deep learning has moved beyond perception tasks like recognising images or speech into the domain of complex engineering design and optimisation. By treating chip floorplanning as a sequential decision-making problem and applying reinforcement learning, researchers have shown that neural networks can learn expert-level spatial reasoning skills, opening the door to using deep learning in other complex design fields such as circuit design, aircraft engineering, and architecture.